



三相半波可控整流电路



电阻负载

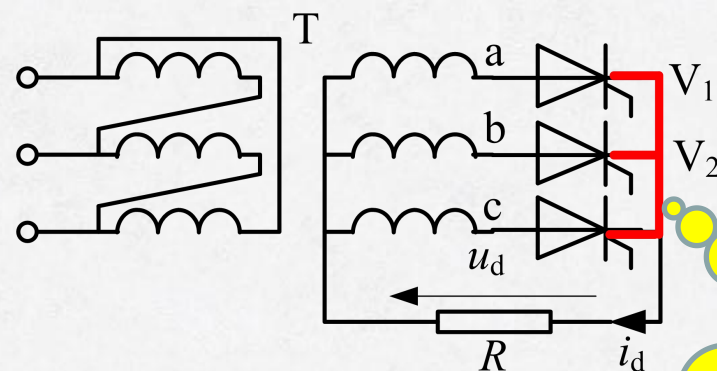


阻感负载

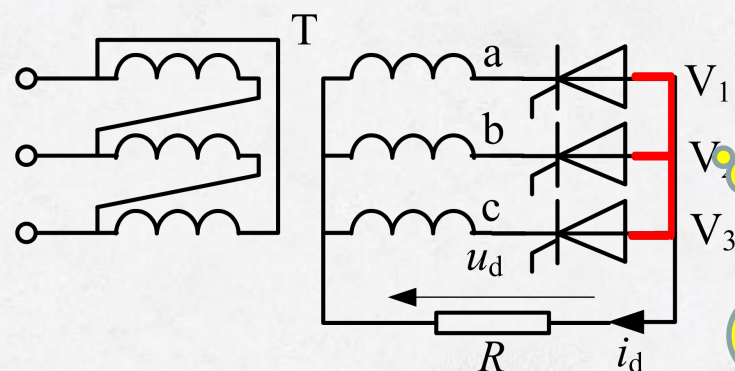
三相半波可控整流电路

■ 电阻性负载

- 变压器二次侧接成星形得到零线接负载，而一次侧接成三角形避免3次谐波流入电网。
- 共阴极接法：三个晶闸管分别接入a、b、c三相电源，其阴极连接在一起。
- 共阳极接法：三个晶闸管分别接入a、b、c三相电源，其阳极连接在一起。



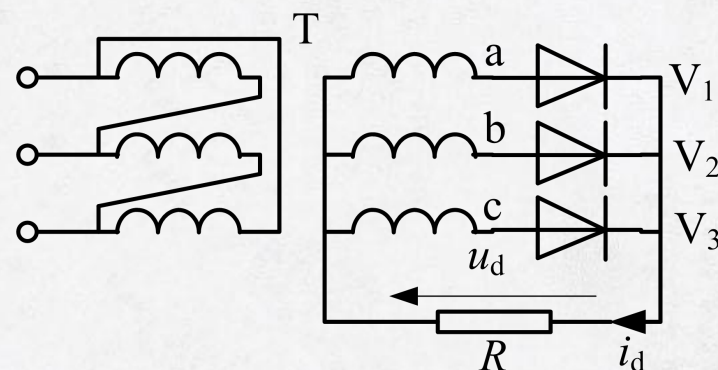
阳极电压最高相的晶闸管导通



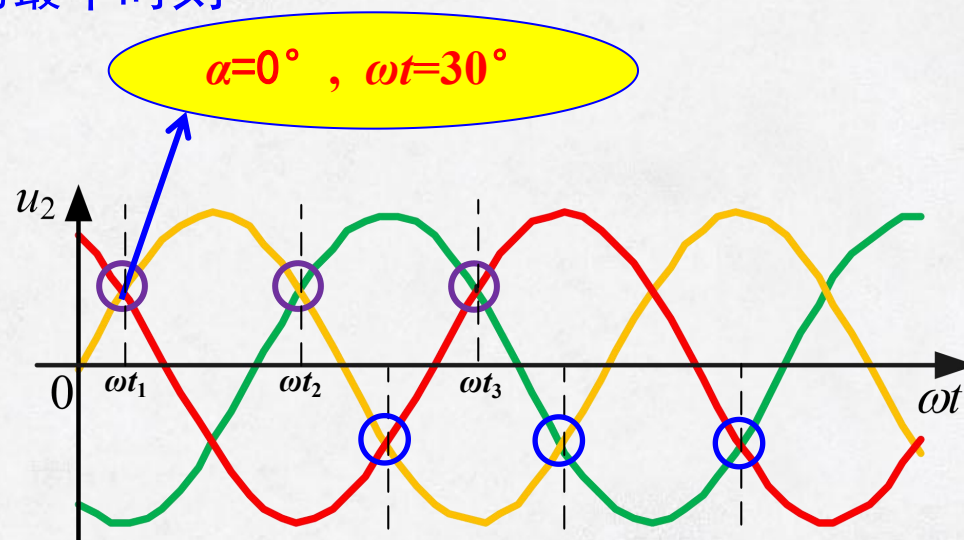
阴极电压最低相的晶闸管导通

三相半波可控整流电路

- 换流：整流管的电流进行交换。对于三相电路，也称换相。
- 自然换流（相）点
- 在三相可控整流电路中，自然换相点即为各晶闸管能触发导通的最早时刻



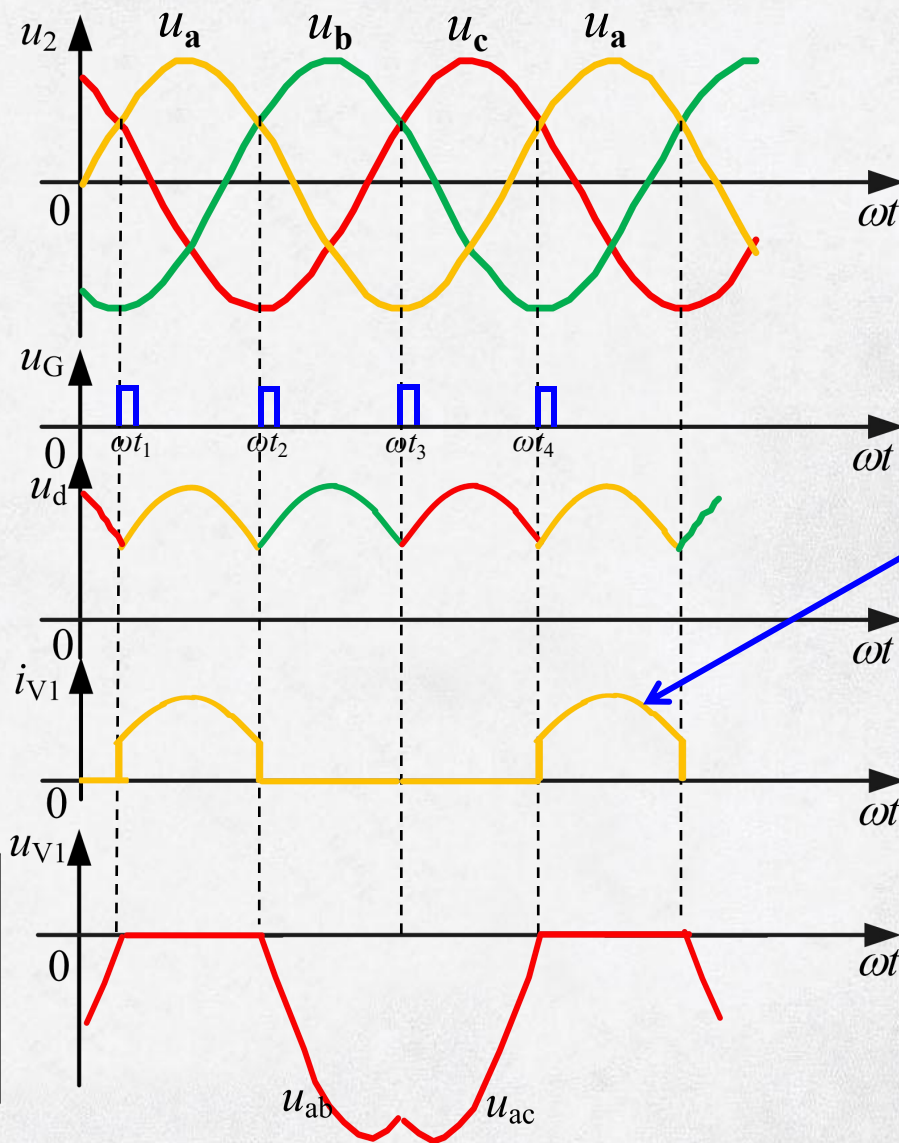
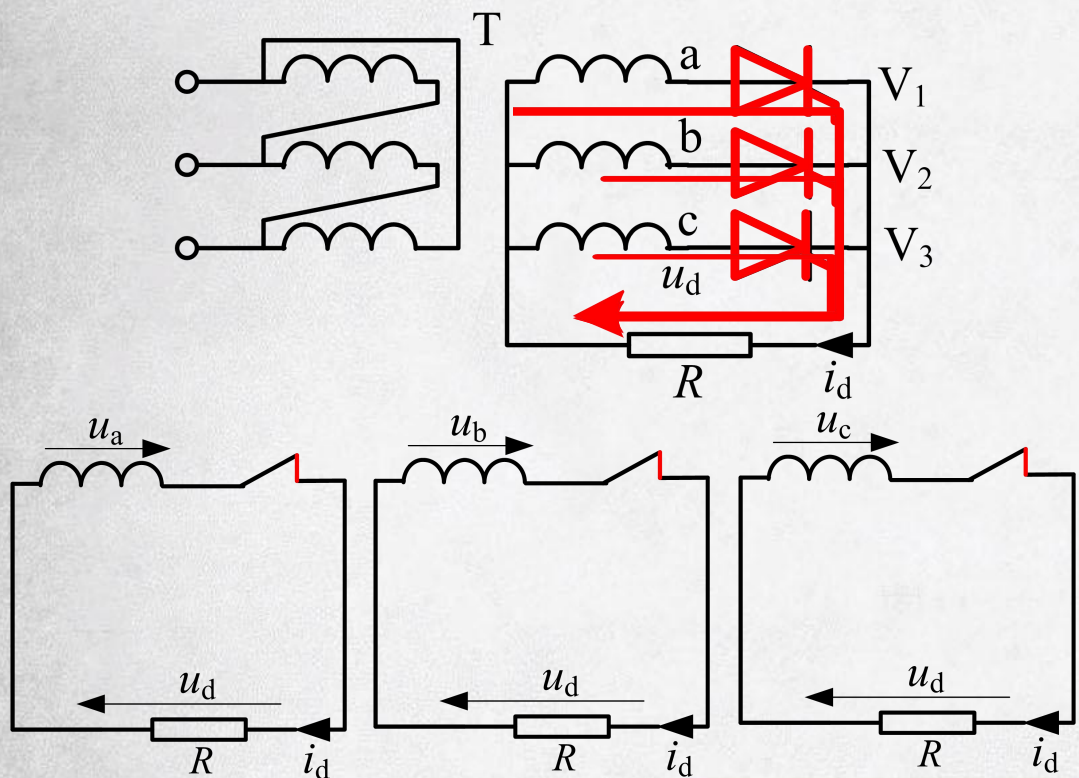
三相半波不可控整流电路



思考：共阳极组的自然换相点和共阴极组相同吗？如果不相同，差多少度？

三相半波可控整流电路

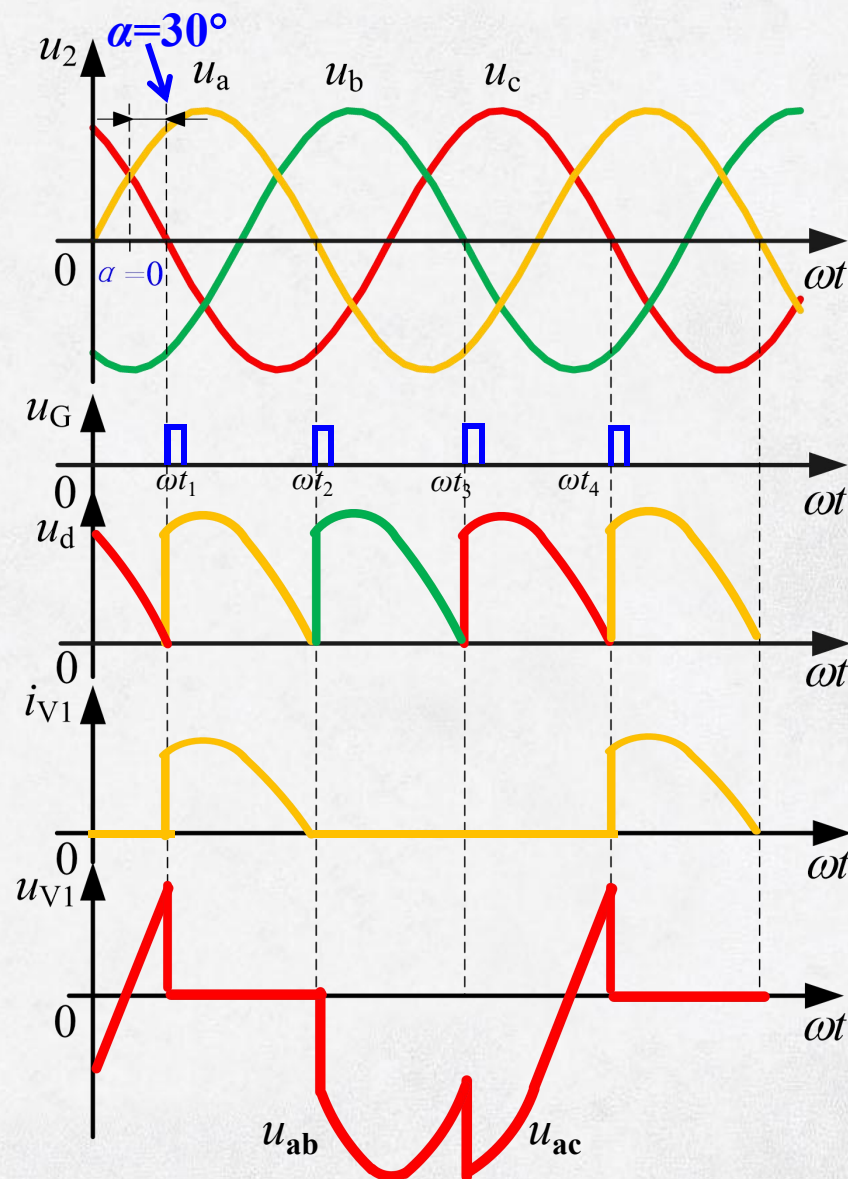
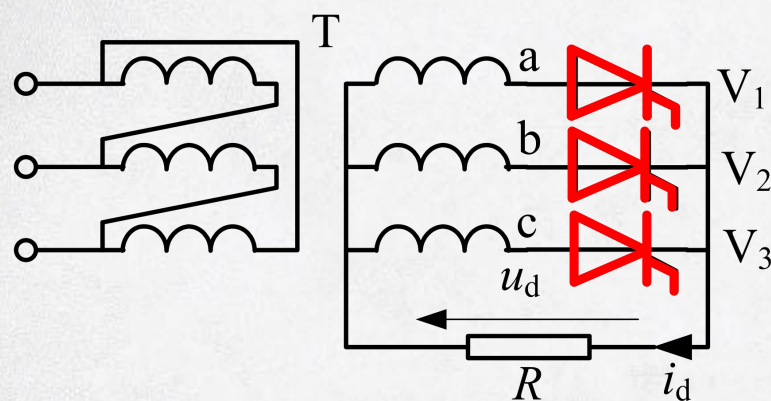
- $\alpha = 0^\circ$
 - 与三只二极管组成的不控整流电路相同



直流分量流过变压器二次绕组，造成铁芯直流磁化

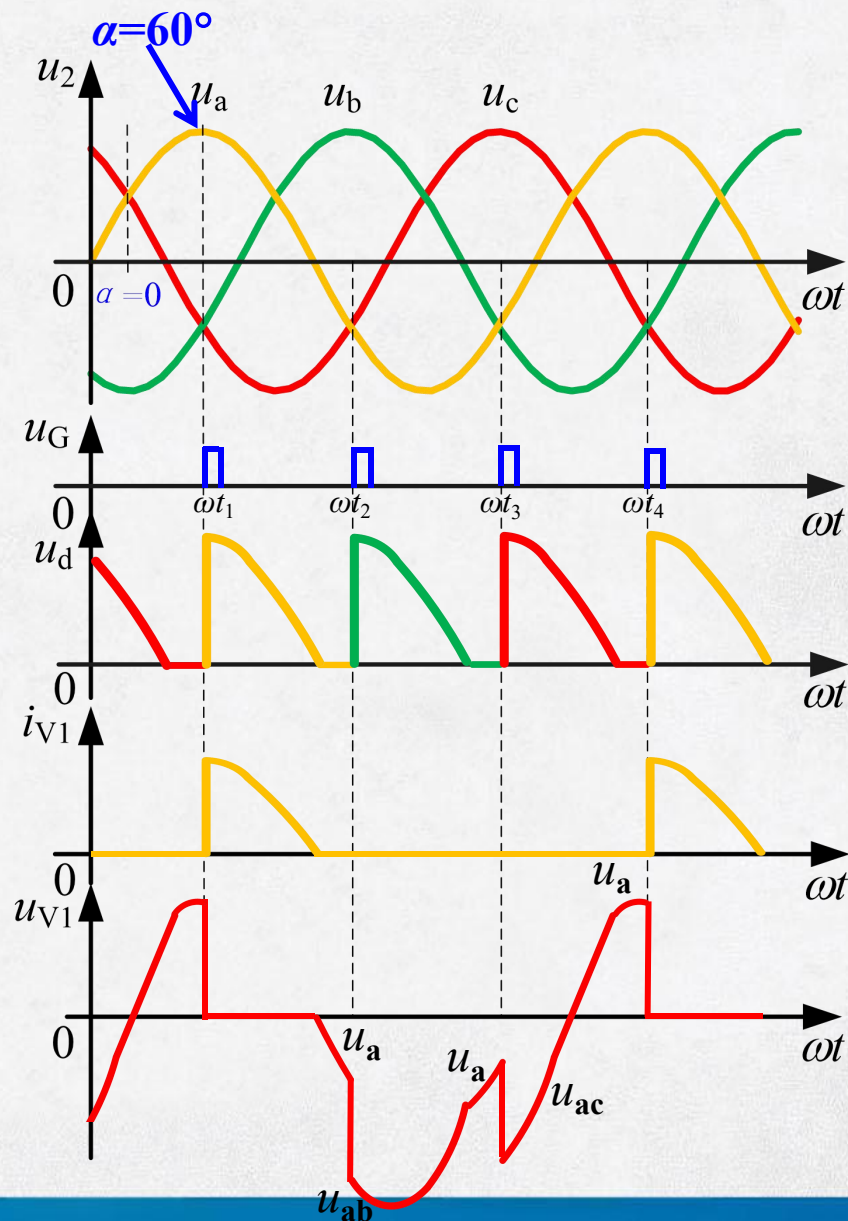
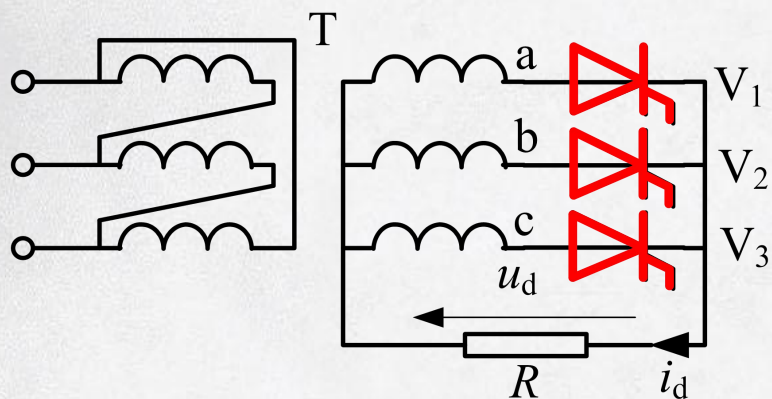
三相半波可控整流电路

- $\alpha = 30^\circ$
 - $\alpha = 30^\circ$ 是电压、电流连续和断续的分界
 - $\theta = 120^\circ$
 - $\alpha > 30^\circ$ 后, 电流断续, $\theta < 120^\circ$



三相半波可控整流电路

- $\alpha = 60^\circ$
 - 电流断续
 - $\theta = 90^\circ$



三相半波可控整流电路

• 数值计算

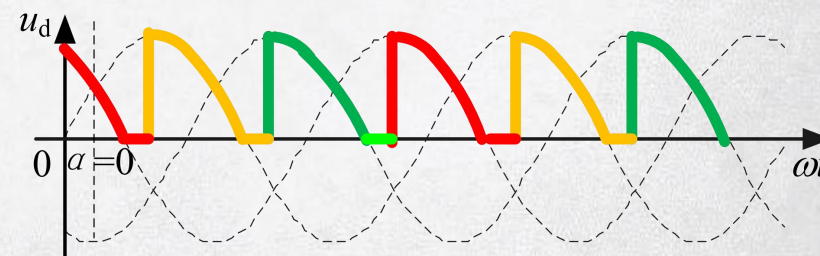
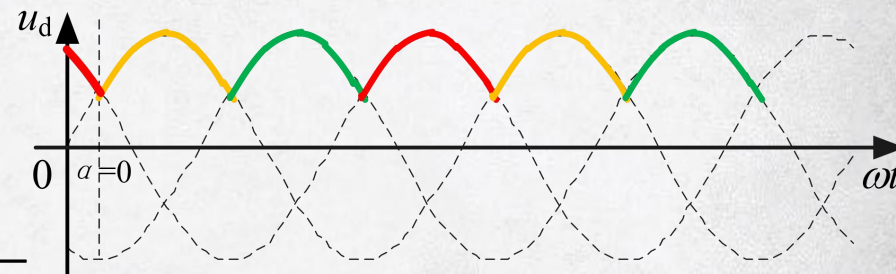
— $\alpha \leq 30^\circ$ 时，负载电流连续，输出电压为

$$U_d = \frac{1}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\frac{5\pi}{6}+\alpha} \sqrt{2}U_2 \sin \omega t d(\omega t) = \frac{3\sqrt{6}}{2\pi} U_2 \cos \alpha = 1.17U_2 \cos \alpha$$

— $\alpha > 30^\circ$ 时，负载电流断续，输出电压为

$$U_d = \frac{1}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\pi} \sqrt{2}U_2 \sin \omega t d(\omega t) = \frac{3\sqrt{2}}{2\pi} U_2 \left[1 + \cos\left(\frac{\pi}{6} + \alpha\right) \right] = 0.675U_2 \left[1 + \cos\left(\frac{\pi}{6} + \alpha\right) \right]$$

— 移相范围： $0^\circ - 150^\circ$





三相半波可控整流电路

- 负载电流平均值 $I_d = \frac{U_d}{R}$

- 晶闸管电流有效值

$$I_V = \sqrt{\frac{1}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\frac{5\pi}{6}+\alpha} \left(\frac{\sqrt{2}U_2 \sin \omega t}{R} \right)^2 d(\omega t)} = \frac{U_2}{R} \sqrt{\frac{1}{2\pi} \left(\frac{2\pi}{3} + \frac{\sqrt{3}}{2} \cos 2\alpha \right)}$$

$$\alpha \leq \pi / 6$$

$$I_V = \sqrt{\frac{1}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\pi} \left(\frac{\sqrt{2}U_2 \sin \omega t}{R} \right)^2 d(\omega t)} = \frac{U_2}{R} \sqrt{\frac{1}{2\pi} \left(\frac{5\pi}{6} - \alpha + \frac{\sqrt{3}}{4} \cos 2\alpha + \frac{1}{4} \sin 2\alpha \right)}$$

$$\alpha > \pi / 6$$

- 晶闸管承受的最大反向电压 $U_{RM} = \sqrt{6}U_2 = 2.45U_2$

- 晶闸管承受的最大正向电压 $U_{FM} = \sqrt{2}U_2$

三相半波可控整流电路

■ 阻感性负载

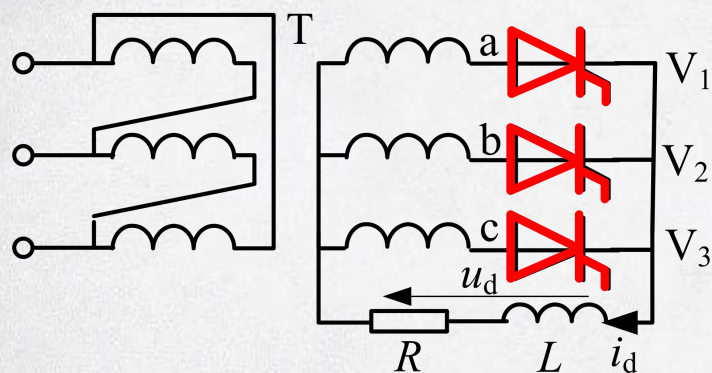
– 电感足够大，使电流连续

• $\alpha = 60^\circ$

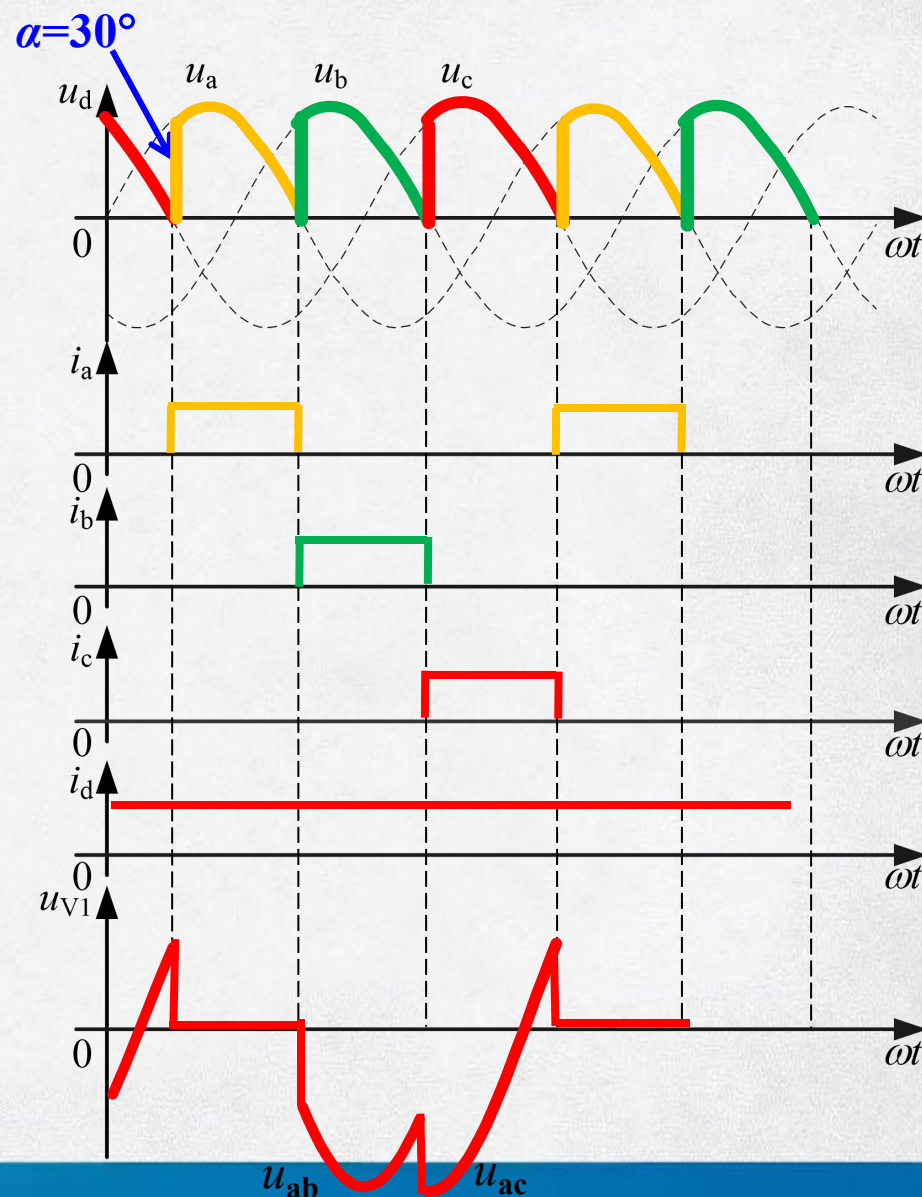
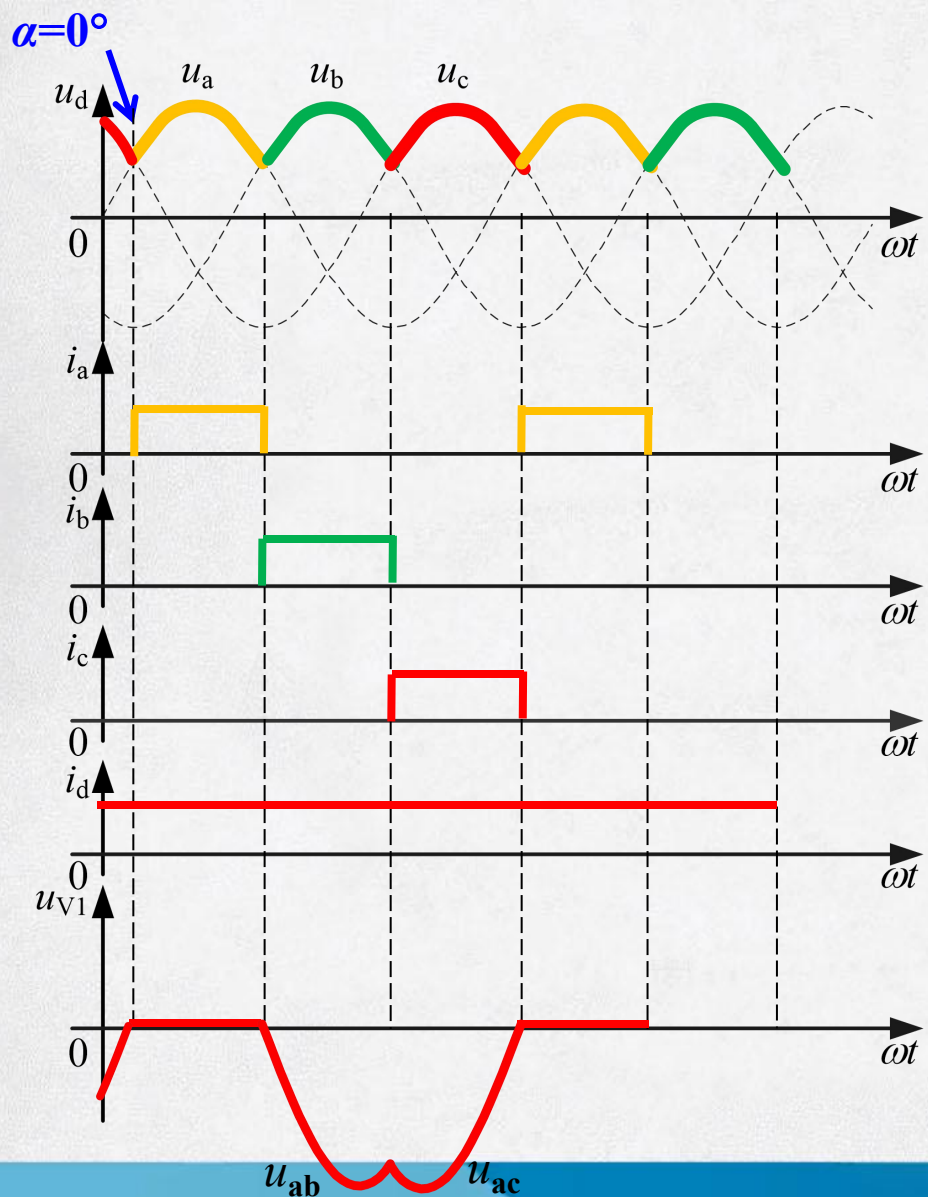
$\alpha > 30^\circ$ 后负载电流波形仍连续

导通角 $\theta = 120^\circ$

理想情况下负载电流波形为直线



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三相半波可控整流电路

数值计算

- 负载电压 $U_d = 1.17U_2 \cos \alpha$

- 移相范围 $0^\circ \sim 90^\circ$

- 负载电流 $I_d = \frac{U_d}{R} = 1.17 \frac{U_2}{R} \cos \alpha$

- 晶闸管电流有效值 $I_V = \frac{1}{\sqrt{3}} I_d$

- 变压器二次侧及晶闸管电流有效值 $I_2 = I_V = \frac{1}{\sqrt{3}} I_d$

- 晶闸管最大正、反向电压峰值: $U_{FM} = U_{RM} = 2.45U_2$

- 变压器二次电流中含有直流分量，应用较少

